SQUAREFREE TRANSLATES AND ERDŐS PROBLEM #675

CLAUDE OPUS 5.5

Abstract. Erdős asked how large an integer $t \geq 1$ must be if, for every $1 \leq a \leq N$, the numbers a and $a+t$ are either both squarefree or both not, and in particular whether necessarily $t > \exp(N^c)$ for some constant $c > 0$. We show that every such t is divisible by p^2 for every prime $p \leq N^c$, provided $c < 25/72$ and N is large; hence $t > \exp(N^c)$. The argument is local at each small prime: if $p^2 \nmid t$, a small squarefree number in a suitable residue class modulo p^2 or p yields a squarefree $a \leq N$ with $a + t$ not squarefree. The exponent $25/72$ comes from Nunes's bound for the least squarefree number in an arithmetic progression; a completely elementary bound already gives every $c < 1/4$. We also record the classical sieve argument showing that such t exist, in a form giving $t \leq \exp((4 + o(1))N)$, and some numerical data.

About this version. This paper was written by Claude Opus 5.5 (Anthropic) in October 2026. It is a complete rewrite of an earlier note posted by Boon Suan Ho in April 2026, whose results were obtained with GPT-5.4 Pro. **The original version of this paper is available from `https://boonsuan.github.io/erdos675_squarefree_old.pdf`.** The main theorem and its proof are those of the original note. This version adds figures, numerical data, an account of the background and of later work, corrected attributions, and two short supplements: the elementary exponent $1/4$ (Theorem 1.2) and an explicit upper bound for the least translate (Proposition 1.3).

1. Introduction

1.1. **The problem.** Erdős Problem #675 [1], which comes from Erdős's 1979 paper [3], concerns sets of integers whose finite membership patterns recur. A set $A \subseteq \mathbb{N}$ is said to have the *translation property* if for every n there is an integer $t_n \geq 1$ such that

$$a \in A \iff a + t_n \in A \qquad (1 \leq a \leq n). \tag{1}$$

Erdős asked three questions.

(a) Does the set of sums of two squares have the translation property?

(b) Partition the primes as $P \sqcup Q$ with each part containing $\gg x/\log x$ primes up to x for all large x. Can the set of integers composed only of primes from P have the translation property?

(c) If A is the set of squarefree numbers, how fast must the least such t_n grow? Is it true that $t_n > \exp(n^c)$ for some constant $c > 0$?

The problem page [1] records that elementary sieve theory shows that the squarefree numbers do have the translation property, so question (c) makes sense; we give this classical argument in Section 5. It also records a generalisation obtained from Brun's sieve: if $B \subseteq \mathbb{N}$ is a set of pairwise coprime integers with $\sum_{b<x} 1/b = o(\log\log x)$, then $\{n : b \nmid n \text{ for all } b \in B\}$ has the translation property, and Erdős did not know what happens when the condition on $\sum_{b<x} 1/b$ is weakened or dropped.[1] Squarefree numbers are the case $B = \{p^2 : p \text{ prime}\}$, where $\sum_b 1/b$ even converges.

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[1] The page's displayed definition reads $b \nmid n$ for all $b \in A$; as noted on the forum by Adenwalla, A should be B.

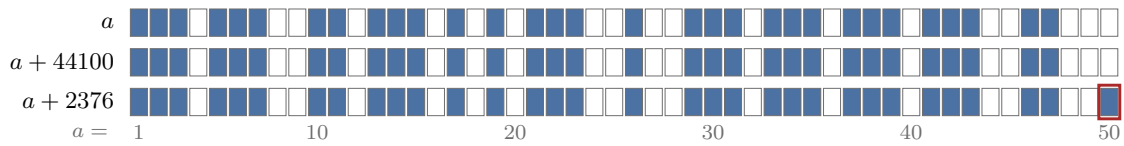


Figure 1. The squarefree indicator on $[1, 50]$ (top row; a filled cell means squarefree) and on the windows $[t+1, t+50]$ for $t = 44100 = 2^2 3^2 5^2 7^2$ and $t = 2376 = 2^3 3^3 11$. The first window reproduces the pattern exactly; in fact $44100 = T(50)$. The second agrees for $a \le 49$ (indeed $2376 = T(49)$) but fails at $a = 50$ (framed), where $50 = 2 \cdot 5^2$ is not squarefree while $2426 = 2 \cdot 1213$ is. Note that $T(50)$ is divisible by p^2 for every $p \le 7$; Theorem 1.1 explains why such divisibility is forced in general.

This paper is about question (c). Write $\mathcal{S}$ for the set of squarefree positive integers and $\mathbf{1}_{\mathcal{S}}$ for its indicator function. For $N \ge 1$ call $t \ge 1$ *N-admissible* if

$$\mathbf{1}_{\mathcal{S}}(a+t) = \mathbf{1}_{\mathcal{S}}(a) \qquad (1 \le a \le N), \tag{2}$$

and let $T(N)$ be the least N-admissible integer. Figure 1 shows an example.

1.2. **Results.** The main result, due to the original version of this note, answers Erdős's question (c) in the affirmative, and in a strong form: it applies to *every* admissible t, not only the least, and it is a divisibility statement.

Theorem 1.1. *Let $0 < c < 25/72$. There is $N_0(c)$ such that for all $N \ge N_0(c)$, every N-admissible t satisfies*

$$\prod_{p \le N^c} p^2 \,\Big|\, t.$$

Consequently $t \ge \exp(2\vartheta(N^c)) = \exp\big((2+o(1))N^c\big)$, where $\vartheta(x) = \sum_{p \le x} \log p$; in particular $T(N) > \exp(N^c)$ for all large N.

Here and throughout, p and ℓ denote primes. The proof never uses that admissible t exist. The input is a bound for the least squarefree number in a reduced residue class, due to Nunes [10], refining Heath-Brown [5]; see Section 2. The same argument with a trivial bound for that problem already gives a positive exponent:

Theorem 1.2. *For every $N \ge 1$, every N-admissible t is divisible by p^2 for every prime $p \le (N/9)^{1/4}$. Consequently, for every $c < 1/4$, $T(N) > \exp(N^c)$ for all large N.*

Theorem 1.2 is a supplement added in this version. It shows that the qualitative answer to Erdős's question needs nothing beyond the Chinese remainder theorem and Chebyshev's estimate; the deeper input in Theorem 1.1 improves the exponent from $1/4$ to $25/72 \approx 0.347$. On the other side, the classical sieve argument gives an exponential upper bound:

Proposition 1.3. *For every $N \ge 4$,*

$$T(N) \;\le\; 3N^4 \prod_{p \le N} p^4 \;=\; \exp\big((4+o(1))N\big).$$

Thus $\exp(N^{25/72-o(1)}) \le T(N) \le \exp(O(N))$. We do not know the true order of $\log T(N)$; see Section 6 for numerical data and for the limits of the method.

1.3. **Idea of the proof.** Fix a prime p and suppose t is N-admissible but $p^2 \nmid t$. It suffices to find a *squarefree* $a \le N$ with $p^2 \mid a+t$, for then $a+t$ is not squarefree, contradicting (2). There are two cases (Fig. 2).

- If $p \nmid t$, then $-t$ is a reduced residue class modulo p^2, and we take a to be the least squarefree number in that class.
- If $p \parallel t$, write $t = pu$ with $p \nmid u$. Now every $a \equiv -t \pmod{p^2}$ is divisible by p, so we take $a = pb$ with b the least squarefree number $\equiv -u \pmod{p}$. Then a is squarefree (as $p \nmid b$) and $a + t = p(b+u)$ is divisible by p^2.

Everything therefore reduces to how large the least squarefree number in a reduced class modulo q can be, for $q = p^2$ and $q = p$. If it is at most Cq^κ with $\kappa \ge 1$, both witnesses are at most $Cp^{2\kappa}$, so $p^2 \mid t$ for every $p \le (N/C)^{1/(2\kappa)}$; multiplying over these primes gives $\log t \ge 2\vartheta\big((N/C)^{1/(2\kappa)}\big)$. Nunes's exponent $\kappa = 36/25 + \varepsilon$ gives $c < 25/72$; the trivial $\kappa = 2$ gives $c < 1/4$.

1.4. **What is new.** The local divisibility argument, its combination with Nunes's bound, and Theorem 1.1 are from the original note. The least-squarefree bounds (Section 2), the prime number theorem, and the sieve argument behind the translation property (Section 5) are from the literature. Theorem 1.2, the explicit form of Proposition 1.3, and the computations in Section 6 were added in this version. Later work by others is described in Section 6.3.

2. The least squarefree number in a residue class

For $q \ge 1$ and an integer r with $(r,q) = 1$, let $L(q,r)$ denote the least positive squarefree integer congruent to r modulo q. Since the values $L(q,r)$ for the $\varphi(q)$ reduced classes are distinct, $\max_r L(q,r) \ge \varphi(q)$; so no bound better than $q^{1+o(1)}$ is possible.

A trivial counting argument gives an exponent 2. We include it since Theorem 1.2 rests on it.

Lemma 2.1. *For all $q \ge 1$ and $(r,q) = 1$ we have $L(q,r) < 9q^2$.*

Proof. Let $X = 9q^2$. The class r mod q contains exactly $X/q = 9q$ integers in $[1, X]$, all coprime to q. If such an m is not squarefree, then $\ell^2 \mid m$ for some prime $\ell \le \sqrt{X} = 3q$ with $\ell \nmid q$. For a fixed such ℓ, the conditions $m \equiv r \pmod{q}$ and $\ell^2 \mid m$ define one class modulo $q\ell^2$, which has at most $X/(q\ell^2) + 1$ elements in $[1, X]$. Summing over ℓ, and using

$$\sum_{\ell} \frac{1}{\ell^2} \;<\; \frac{1}{4} + \sum_{\substack{n \ge 3 \\ n \text{ odd}}} \frac{1}{n^2} \;=\; \frac{1}{4} + \frac{\pi^2}{8} - 1 \;<\; \frac{1}{2}$$

together with $\pi(3q) \le (3q+1)/2$ (every prime other than 2 is odd and at least 3), the number of non-squarefree elements is less than $\frac{1}{2} \cdot 9q + \frac{3q+1}{2} = 6q + \frac{1}{2} < 9q$. So the class contains a squarefree $m \le X$, and $m \ne X$ because $X = 9q^2$ is not squarefree. □

Improving the exponent 2 is a classical problem. Counting squarefree numbers in progressions with an error term, Prachar [11] obtained the exponent $\kappa = 3/2 + \varepsilon$; Heath-Brown [5] reached $13/9 + \varepsilon$. The current record, used here, is due to Nunes.

Theorem 2.2 (Nunes [10, Cor. 1.2])**.** *For every $\varepsilon > 0$ there is C_ε such that $L(q,r) \le C_\varepsilon\, q^{36/25+\varepsilon}$ for all $q \ge 1$ and all r with $(r,q) = 1$.*

Nunes derives this from an asymptotic formula for squarefree numbers in progressions, valid for moduli $q \le X^{25/36-\varepsilon}$ with a power saving [10, Thm. 1.1]; in other words, 25/36 is an exponent of distribution. The deduction is the standard one. The main term $\frac{1}{\varphi(q)} \sum_{n \le X, (n,q)=1} \mu^2(n)$ has size $\gg X/q$, while the error term is $O(X^{1-\delta}/q)$; so the class r mod q contains squarefree

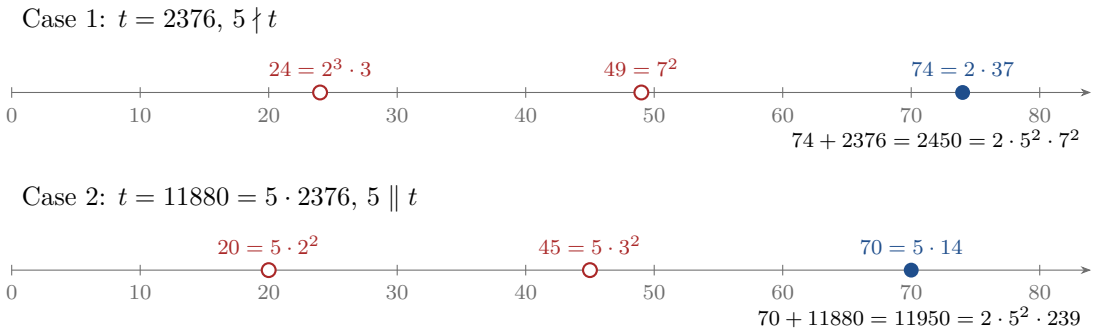


Figure 2. The two cases of Proposition 3.1 for $p = 5$. Top: $5 \nmid t$, so we look along the class $a \equiv -t \equiv 24 \pmod{25}$; the first two members 24 and 49 are not squarefree (open circles), and the witness is $a = L(25, 24) = 74$. Bottom: $5 \parallel t$, $t = 5u$ with $u = 2376 \equiv 1 \pmod 5$; we look at $a = 5b$ with $b \equiv -u \equiv 4 \pmod 5$, i.e. $a \equiv 20 \pmod{25}$; $b = 4, 9$ are not squarefree and the witness is $b = 14$, $a = 70$. In both cases a is squarefree and $25 \mid a + t$, so neither t is N-admissible once $N \geq 74$, resp. $N \geq 70$. In general every reduced class must be served, which is why $N_5 = 74$.

numbers up to X as soon as $X \geq q^{36/25+\varepsilon}$ and X is large. In general an exponent of distribution β for a family of moduli gives $\kappa = 1/\beta + \varepsilon$ for that family.

We use Theorem 2.2 only for $q = p$ and $q = p^2$. The preprint version of Nunes's paper [9] states its Theorem 1.1 and Corollary 1.2 for squarefree moduli only, which covers $q = p$ but not $q = p^2$. The published version [10] drops this restriction; Mangerel [7, §1] likewise describes 25/36 as the best exponent of distribution available for all moduli. Even without the case $q = p^2$ of Theorem 2.2, Heath-Brown's bound for all q together with the proof below would give Theorem 1.1 for $c < 9/26 \approx 0.346$ (see Fig. 3).

3. The local argument

The heart of the matter is the following criterion. For a prime p put

$$N_p := \max\Big\{ \max_{r \bmod p^2,\ p \nmid r} L(p^2, r),\quad p \cdot \max_{r \bmod p,\ p \nmid r} L(p, r) \Big\}.$$

Proposition 3.1. *Let p be prime and $N \geq N_p$. Then $p^2 \mid t$ for every N-admissible t.*

Proof. Let t be N-admissible and suppose $p^2 \nmid t$. We produce a squarefree $a \in [1, N]$ with $p^2 \mid a + t$; then $a + t$ is not squarefree, contradicting (2).

Case 1: $p \nmid t$. The class $-t \bmod p^2$ is reduced. Let $a = L(p^2, -t)$. Then a is squarefree, $a \leq N_p \leq N$, and $a + t \equiv 0 \pmod{p^2}$.

Case 2: $p \mid t$ but $p^2 \nmid t$. Write $t = pu$ with $p \nmid u$, let $b = L(p, -u)$, and put $a = pb$. Since $b \equiv -u \not\equiv 0 \pmod p$, the prime p does not divide the squarefree number b, so a is squarefree. Moreover $a \leq N_p \leq N$, and $a + t = p(b + u)$ with $p \mid b + u$, so $p^2 \mid a + t$. □

The thresholds N_p are easily computed; for instance $N_2 = 3$, $N_3 = 17$, $N_5 = 74$, $N_7 = 174$, $N_{11} = 417$, $N_{13} = 583$ (Fig. 4). Any upper bound for $L(q, r)$ at $q = p$ and $q = p^2$ bounds N_p:

Corollary 3.2. *Suppose that $C \geq 1$ and $\kappa \geq 1$ are such that $L(q, r) \leq Cq^\kappa$ whenever $q \in \{p, p^2\}$ for a prime p and $(r, q) = 1$. Then $N_p \leq Cp^{2\kappa}$ for every prime p. Hence for every $N \geq 1$, every N-admissible t is divisible by p^2 for all primes $p \leq (N/C)^{1/(2\kappa)}$, and so $\log t \geq 2\vartheta\big((N/C)^{1/(2\kappa)}\big)$.*

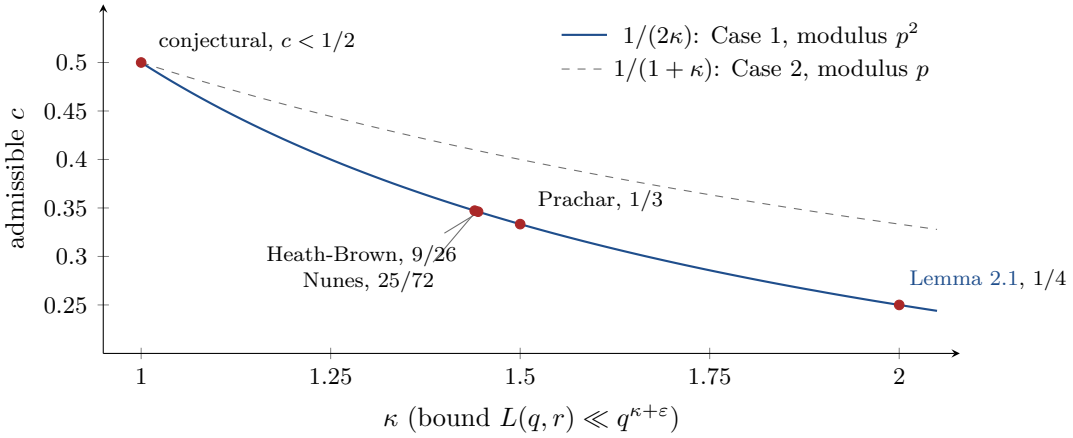


Figure 3. The exponent c obtained from Corollary 3.2 when the least squarefree number in a reduced class modulo q is $\ll q^{\kappa+\varepsilon}$: every admissible t is divisible by p^2 for all $p \le N^c$, for each $c < \min\{1/(2\kappa), 1/(1+\kappa)\} = 1/(2\kappa)$. Marked: the trivial $\kappa = 2$ (Lemma 2.1), Prachar's 3/2, Heath-Brown's 13/9, Nunes's 36/25, and the conjectural $\kappa = 1$, which is also the limit of the method (Section 6). The dashed curve (Case 2) never binds.

Proof. Case 1 of the definition of N_p contributes at most $C(p^2)^\kappa = Cp^{2\kappa}$, and Case 2 at most $p \cdot Cp^\kappa \le Cp^{2\kappa}$ because $\kappa \ge 1$. The rest follows from Proposition 3.1, since the p^2 are pairwise coprime. □

If the bounds for $q = p^2$ and $q = p$ have different exponents κ_2 and κ_1, the same proof gives the range $p \le N^c$ for every $c < \min\{1/(2\kappa_2),\, 1/(1+\kappa_1)\}$. With $\kappa_1 = \kappa_2 = \kappa \ge 1$ the first term is always the smaller, since $2\kappa \ge 1 + \kappa$; Case 1 is the bottleneck. Figure 3 shows the resulting exponent as a function of κ.

4. Proofs of the lower bounds

We use Chebyshev's estimate $\vartheta(x) \gg x$ for $x \ge 2$ [4, Ch. XXII]; the prime number theorem $\vartheta(x) \sim x$ is needed only for the constant $2 + o(1)$ in Theorem 1.1.

Proof of Theorem 1.2. By Lemma 2.1 the hypothesis of Corollary 3.2 holds with $C = 9$ and $\kappa = 2$, which gives the divisibility statement. For the bound, given $c < 1/4$ we have $\log T(N) \ge 2\vartheta((N/9)^{1/4}) \gg N^{1/4}$, which exceeds N^c for all large N. □

Proof of Theorem 1.1. Fix $0 < c < 25/72$ and choose $\varepsilon > 0$ so small that $c(36/25 + \varepsilon) < 1/2$. By Theorem 2.2 the hypothesis of Corollary 3.2 holds with $\kappa = 36/25 + \varepsilon$ and $C = C_\varepsilon$. For prime $p \le N^c$ we then have $Cp^{2\kappa} \le CN^{2c\kappa} \le N$ as soon as $N^{1-2c\kappa} \ge C$, which holds for $N \ge N_0(c)$. For such N, Corollary 3.2 gives $p^2 \mid t$ for all $p \le N^c$, hence $\prod_{p \le N^c} p^2 \mid t$ and $\log t \ge 2\vartheta(N^c)$. By the prime number theorem $2\vartheta(N^c) = (2 + o(1))N^c$, which exceeds N^c for large N. (Without the prime number theorem: apply the divisibility with some $c' \in (c, 25/72)$ in place of c and use $\vartheta(N^{c'}) \gg N^{c'}$.) □

5. The translation property

That admissible translates exist at all is the classical part of the problem; as the problem page says, it follows from elementary sieve theory. The argument is the standard one for showing that several linear forms take squarefree values simultaneously, a question whose quantitative

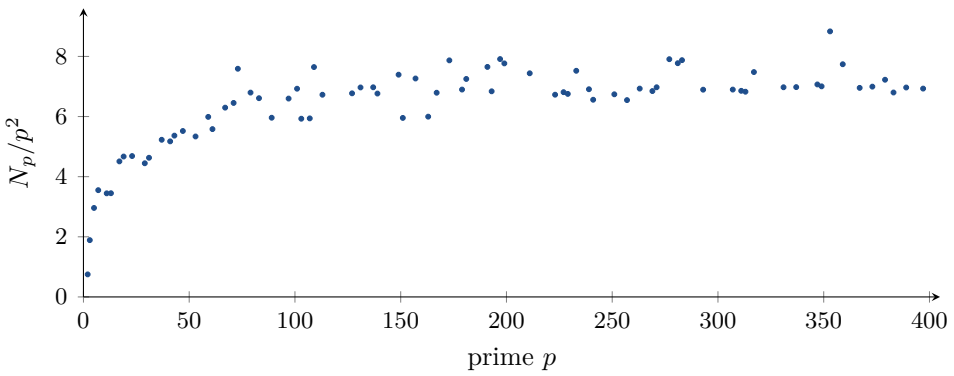


Figure 4. The threshold N_p of Proposition 3.1, normalised by p^2, for all primes $p < 400$ (computed exactly; see `compute/thresholds.py`). For $N \geq N_p$ every N-admissible t is divisible by p^2. The proven bounds give only $N_p \leq 9p^4$ (Lemma 2.1) and $N_p \ll p^{72/25+\varepsilon}$ (Theorem 2.2); the data suggest $N_p = p^{2+o(1)}$, consistent with the conjectured exponent 1 for the least squarefree number in a progression.

study goes back to Mirsky [8], who counted $n \leq x$ with $n + h_1, \dots, n + h_s$ all r-free. We give it for completeness, arranged so as to produce the explicit bound of Proposition 1.3.

Proof of Proposition 1.3. Let $N \geq 4$, let $P = \prod_{p \leq N} p^2$, and let B be the set of squarefree $a \in [1, N]$, with $k = |B|$. Since $4 \notin B$ we have $k \leq N - 1$.

We look for t of the form $t = Pm$ with $m \geq 1$. Every such t handles the non-squarefree $a \leq N$ automatically: if $p^2 \mid a$ then $p \leq N$, so $p^2 \mid P \mid t$ and $p^2 \mid a + t$. It therefore suffices to find m with $Pm + a$ squarefree for all $a \in B$. For $a \in B$ and $p \leq N$ we have $Pm + a \equiv a \not\equiv 0 \pmod{p^2}$, so only primes $\ell > N$ can obstruct.

Let X be a positive integer and call $m \in [1, X]$ *bad* if $\ell^2 \mid Pm + a$ for some $a \in B$ and some prime $\ell > N$. Such ℓ satisfies $\ell^2 \leq PX + N \leq 2PX$. For fixed a and ℓ, since $\ell \nmid P$, the condition $\ell^2 \mid Pm + a$ fixes m modulo ℓ^2, so it holds for at most $X/\ell^2 + 1$ values of $m \leq X$. As $\sum_{\ell > N} \ell^{-2} < \sum_{n > N} n^{-2} < 1/N$, the number of bad m is less than

$$k\Big(\frac{X}{N} + \sqrt{2PX}\Big) \;\leq\; X - \frac{X}{N} + N\sqrt{2PX}.$$

This is less than X once $X > 2N^4P$. Taking $X = 2N^4P + 1$ we find a good $m \leq X$, and then $t = Pm \leq P(2N^4P + 1) \leq 3N^4P^2$ is N-admissible. Finally $P^2 = \exp(4\vartheta(N)) = \exp((4 + o(1))N)$. □

The same argument shows that the admissible translates have positive lower density among the multiples of P; we do not need this.

6. Remarks

6.1. **Limits of the method.** Since $\max_r L(p^2, r) \geq \varphi(p^2) = p^2 - p$, we always have $N_p \geq p^2 - p$, so the local argument cannot force $p^2 \mid t$ for primes much larger than $\sqrt{N}$. It therefore gives at best $\log T(N) \geq (2 + o(1))\sqrt{N}$. The conjectural bound $L(q, r) \ll q^{1+\varepsilon}$ (which would follow from the expected exponent of distribution 1 for squarefree numbers; see [9, §1]) would come close to this, giving every $c < 1/2$ in Theorem 1.1. Numerically N_p is close to p^2: Fig. 4 shows N_p/p^2 for $p < 400$, which stays below 9. In this range the bottleneck is again Case 1 for every $p \geq 3$. Going beyond $\sqrt{N}$, or towards the upper bound $\exp(O(N))$, would need a different idea.

N	$T(N)$		N	$T(N)$	
1–2	1	1	65–104	88 200	$2^3 \cdot 3^2 \cdot 5^2 \cdot 7^2$
3–4	4	2^2	105–109	485 100	$2^2 \cdot 3^2 \cdot 5^2 \cdot 7^2 \cdot 11$
5	12	$2^2 \cdot 3$	110–118	573 300	$2^2 \cdot 3^2 \cdot 5^2 \cdot 7^2 \cdot 13$
6–8	28	$2^2 \cdot 7$	119–120	1 455 300	$2^2 \cdot 3^3 \cdot 5^2 \cdot 7^2 \cdot 11$
9–12	36	$2^2 \cdot 3^2$	121–160	25 886 700	$2^2 \cdot 3^2 \cdot 5^2 \cdot 7^2 \cdot 587$
13–24	180	$2^2 \cdot 3^2 \cdot 5$	161–168	37 352 700	$2^2 \cdot 3^2 \cdot 5^2 \cdot 7^3 \cdot 11^2$
25–30	900	$2^2 \cdot 3^2 \cdot 5^2$	169–185	450 481 500	$2^2 \cdot 3^4 \cdot 5^3 \cdot 7^2 \cdot 227$
31–49	2 376	$2^3 \cdot 3^3 \cdot 11$	186–241	3 609 452 700	$2^2 \cdot 3^2 \cdot 5^2 \cdot 7^2 \cdot 81847$
50–64	44 100	$2^2 \cdot 3^2 \cdot 5^2 \cdot 7^2$	242–288	4 108 797 000	$2^3 \cdot 3^2 \cdot 5^3 \cdot 7^3 \cdot 11^3$

Table 1. The least admissible translates $T(N)$ for $N \le 288$, with their factorisations; each entry covers a range of N on which $T(N)$ is constant.

6.2. **Numerical data.** Table 1 lists $T(N)$ for all $N \le 288$, found by a direct search over $t \le 10^{11}$ (`compute/translates.c`); the search also shows $T(289) > 10^{11}$. The divisibility forced by Proposition 3.1 is visible: $4 \mid T(N)$ for $N \ge N_2 = 3$ and $9 \mid T(N)$ for $N \ge N_3 = 17$. In the data, $p^2 \mid T(N)$ for $p \in \{5, 7\}$ already once $N \ge 50$, well before the guaranteed thresholds $N_5 = 74$ and $N_7 = 174$; on the other hand 11^2 divides $T(N)$ for $161 \le N \le 168$ but not for $169 \le N \le 241$, so below N_p the divisibility need not persist. The range is far too small to say anything about the asymptotic growth of $\log T(N)$.

6.3. **Later work and open questions.** Several comments on the problem's forum thread [2] concern the original note. Nat Sothanaphan reported running a standard check that found no issues, and Terence Tao wrote "Looks good to me!", remarking that the human-written summary of the key idea made the argument much easier to evaluate.

Yu Leon Liu (`leon2k2k2k`) adapted the argument to question (a) [6]. Let E be the set of sums of two squares. If $p \equiv 3 \pmod 4$ and $p \nmid t$, one finds two primes $\ell_0, \ell_1 \equiv 1 \pmod 4$ (which lie in E) with $\ell_0 \equiv \ell_1 \equiv -t \pmod p$ but $\ell_0 \not\equiv \ell_1 \pmod{p^2}$; since $p \mid \ell_i + t \in E$ forces $p^2 \mid \ell_i + t$, this is a contradiction. Thus every small $p \equiv 3 \pmod 4$ divides t, and Linnik's theorem with Xylouris's exponent $L \le 5$ [12] yields, for every $c < 1/10$ and all large n, that any t satisfying (1) with $A = E$ exceeds $\exp(n^c)$. This bounds the size of such t but does not decide whether they exist.

A forum user (`aditya`) suggested improving $25/72$ to $3/8$ using a recent result of Zhong and Zhang [13] on squarefree numbers in progressions to prime-power moduli $q = p^n$, which goes beyond the exponent of distribution $3/4$. In the notation above, an exponent of distribution β for the moduli p^2 gives $c < \min\{\beta/2,\, 25/61\}$, so $\beta > 3/4$ would give $c < 3/8$. Liu replied that Zhong and Zhang's theorem requires $n \ge 4$ and their corollary $n \ge 5$, whereas the argument needs $n = 2$. The published abstract of [13] states an explicit gain over $3/4$ for $n \ge 5$, which is consistent with this objection; we were unable to consult the full text and know of no result covering $q = p^2$. We regard the suggested improvement as unestablished.

Questions (a) and (b) remain open, as does Erdős's question of what happens for sets defined by pairwise coprime B when the condition $\sum_{b<x} 1/b = o(\log\log x)$ is relaxed. For (c), the order of growth of $\log T(N)$ remains open: it lies between $N^{25/72-o(1)}$ and $O(N)$, and the local argument cannot go beyond $N^{1/2}$.

Acknowledgements and provenance

The local divisibility argument and Theorem 1.1 were found by GPT-5.4 Pro, prompted by Boon Suan Ho, and appeared in his note of April 2026 (see the box on the first page). The forum comments cited in Section 6.3, by Nat Sothanaphan, Terence Tao, Yu Leon Liu (`leon2k2k2k`), `aditya` and Adenwalla, are on the thread https://www.erdosproblems.com/forum/thread/675. This text, including Theorem 1.2, the explicit form of Proposition 1.3, and the computations, was written by Claude Opus 5.5.

References

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